

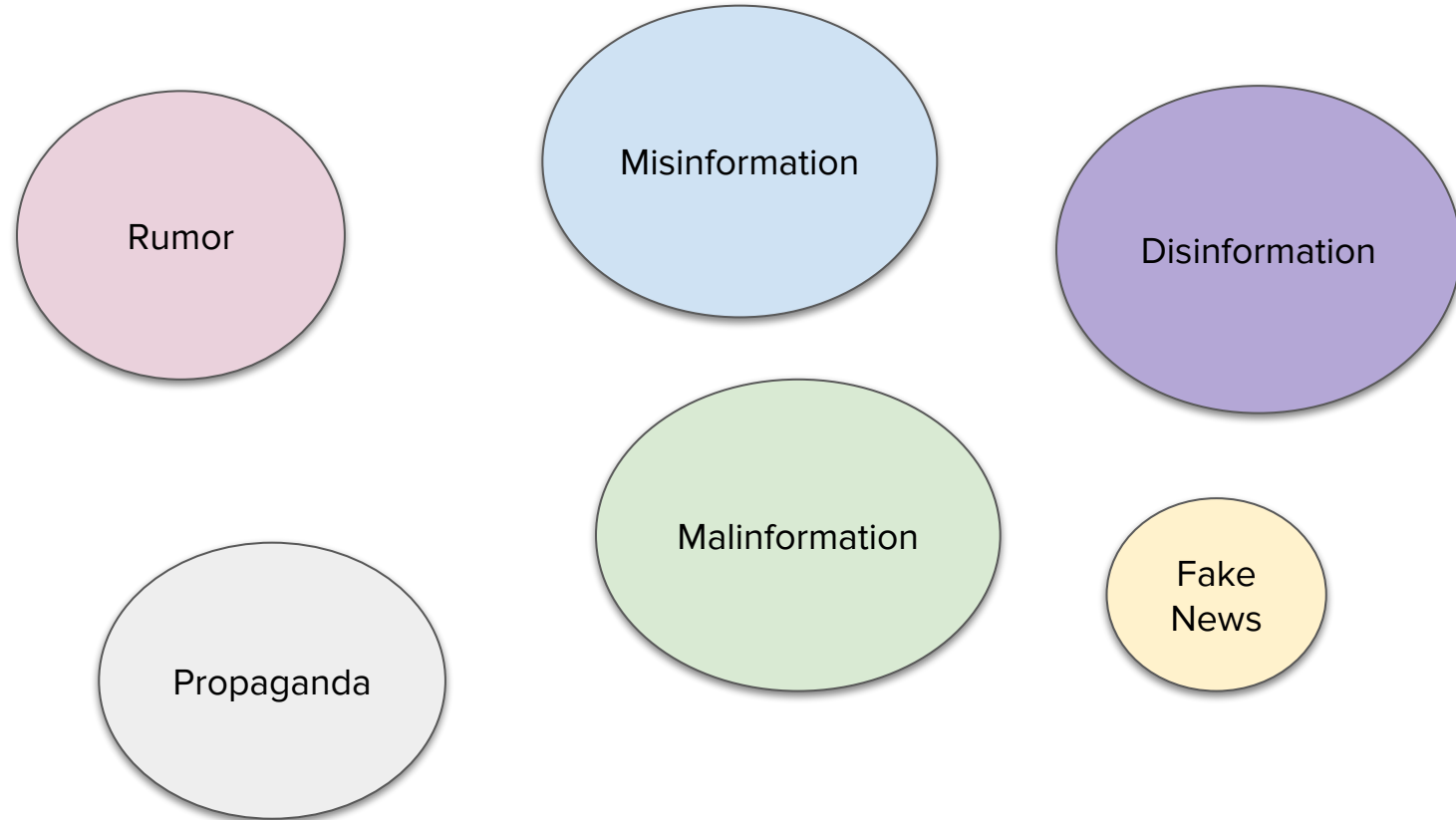
Detecting Fake News and Misinformation in Social Media



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY **DELHI**

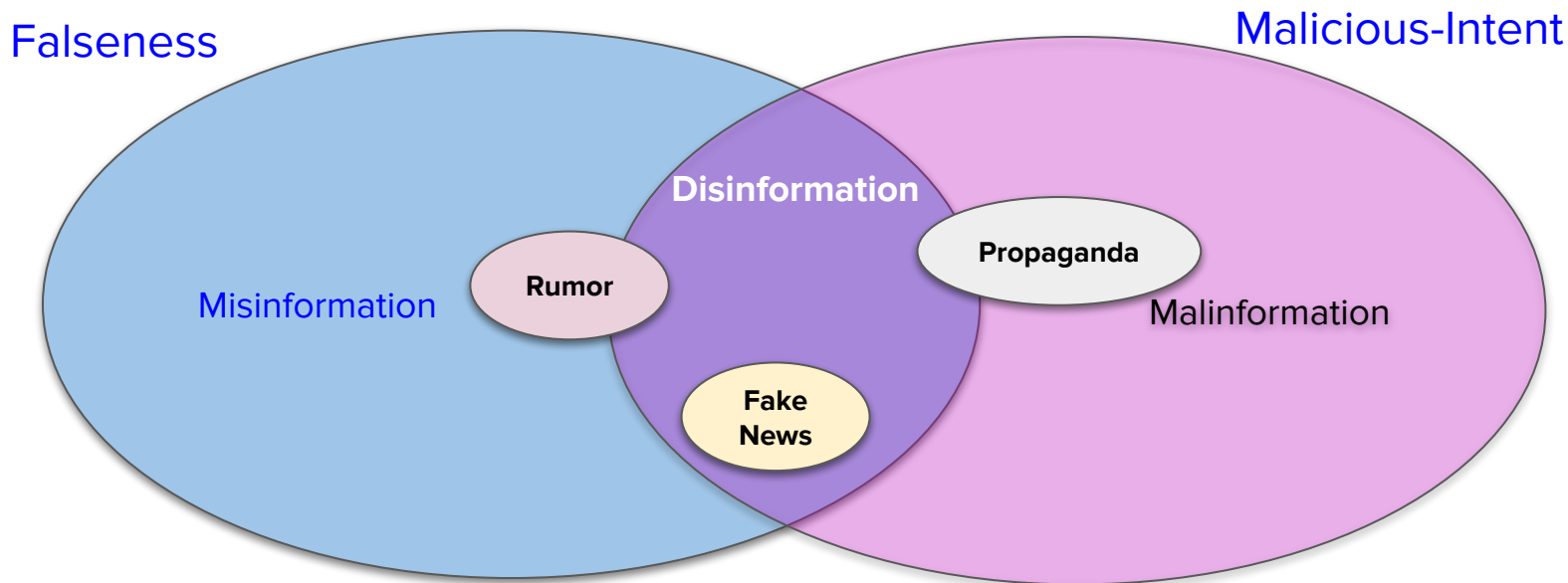


Information Disorders



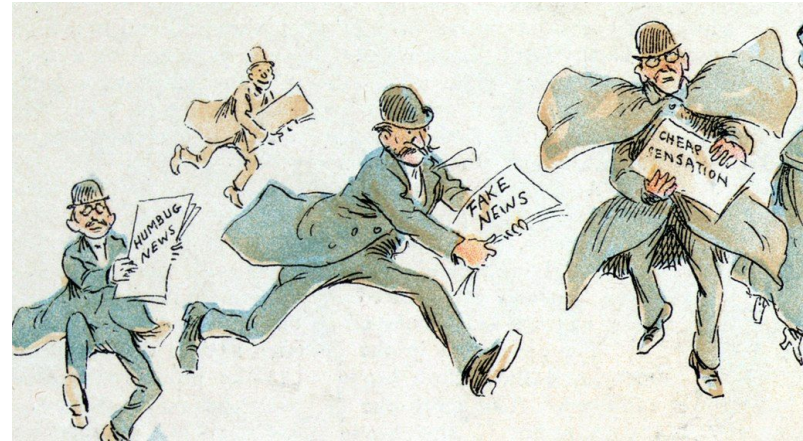
Information Disorders

- Two key ingredients:
 - Veracity: Falseness or Truthfulness
 - Intent: Malicious (intend to harm) or non-malicious



What is *Fake News*?

- The term “fake news” refers to disinformation or false information reported as legitimate news.
- The spread of fake news is often motivated by various factors such as political agendas, financial gain, or simply to provoke reactions and create confusion.



Fake News in History

King on his deathbed

During the Jacobite rebellion, some individuals disloyal to King George II, sought to undermine his rule by printing fake stories that his health was failing.

1835

The Moon hoax

The New York Sun newspaper runs a series of articles claiming that life has been discovered on the Moon. They attributed the discovery to a renowned astronomer, before later admitting the story to be a hoax.

The Zinoviev Letter

Days before the 1924 UK General Election, the Daily Mail publishes a forged letter, purportedly from the Soviet government, suggesting a Labour victory would hasten the radicalisation of the working classes. Labour went on to lose the election.

1924

1989

Hillsborough

In the aftermath of the disaster at Hillsborough football stadium, where 96 supporters died, The Sun newspaper published a series of misleading and fabricated stories, placing much of the blame on Liverpool fans.

US presidential election

Stories proliferated on social media, targeting specific candidates or spreading sensationalist and misleading information to influence public opinion, including Pope Francis had endorsed Donald Trump, false claims about Hillary Clinton's health etc.

2016

Mid-1700

Fighting Fake News

- Manual Verification
- Technology-Based Solutions

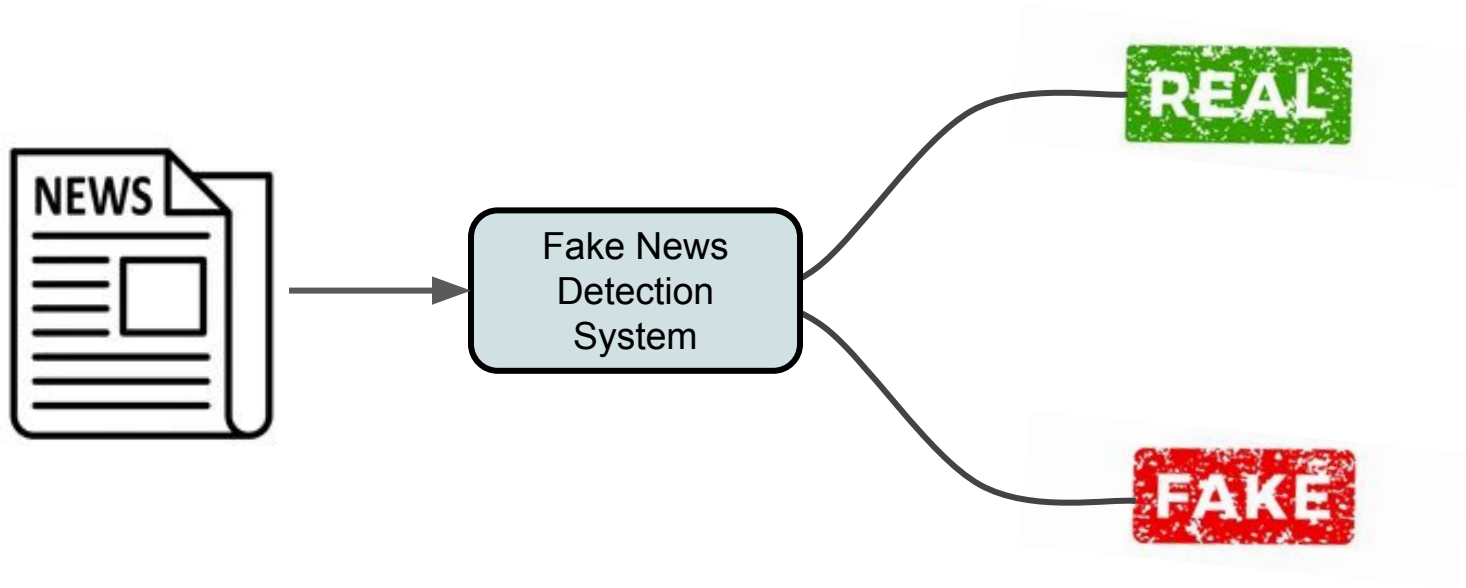
Manual Fact-Checking Organizations



Challenges in Manual Verification

1. High volume of data
 2. Relevance to the world
 3. Limited manpower
 4. Not enough expertise on the topic
 5. Language barrier
 6. Repetitive claims
- so on...

Solution?



Fake News Detection Datasets and competitions

Corpus	Labels	#Examples
BuzzfeedNews	mostly true, mostly false, mixture of true and false	2,282
PHEME	true or false	330
LIAR	pants-fire, false, barely-true, half-true, mostly-true, and true	12,836
FakeNewsNet	fake or real	23,921

CodaLab

[Search Competitions](#) [My Competitions](#) [Help](#) [Sign Up](#) [Sign In](#)



Fake News Detection in Dravidian Languages-
DravidianLangTech@EACL 2024
Organized by DravidianLangTech
Shared Task on Fake News Detection in Dravidian Languages-
DravidianLangTech@EACL 2024

Sep 01, 2023-May 31, 2024
69 participants



CLEF2022 CheckThat! Lab - Task 3
Organized by gautamshahi
Fake News Detection

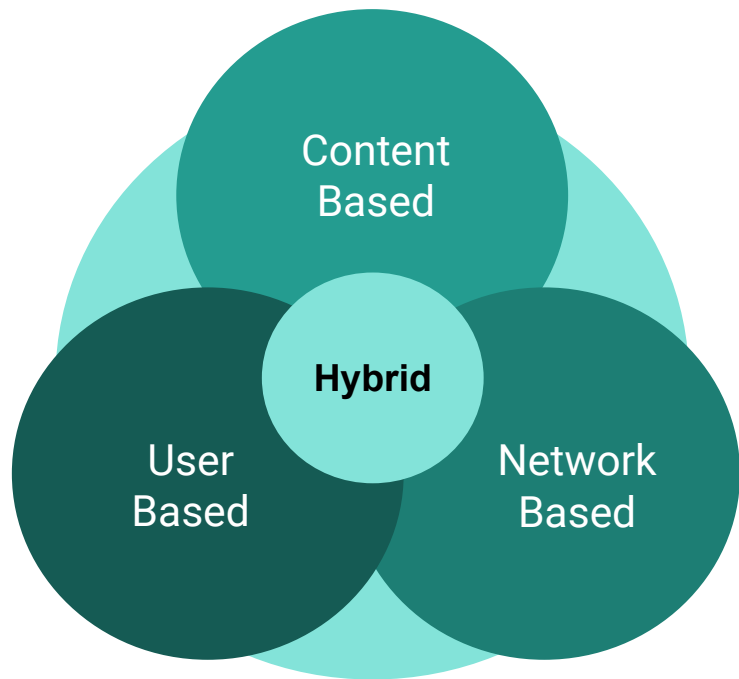
Apr 22, 2022-May 10, 2022
42 participants



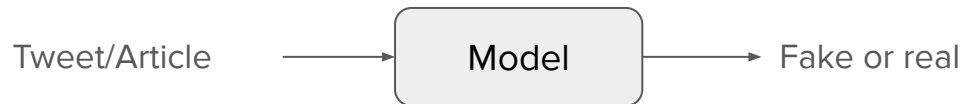
Fake News Detection in Dravidian Languages-
DravidianLangTech@RANLP 2023
Organized by DravidianLangTech
Shared Task on Fake News Detection in Dravidian Languages-
DravidianLangTech@RANLP 2023

Feb 21, 2022-Aug 31, 2023
38 participants

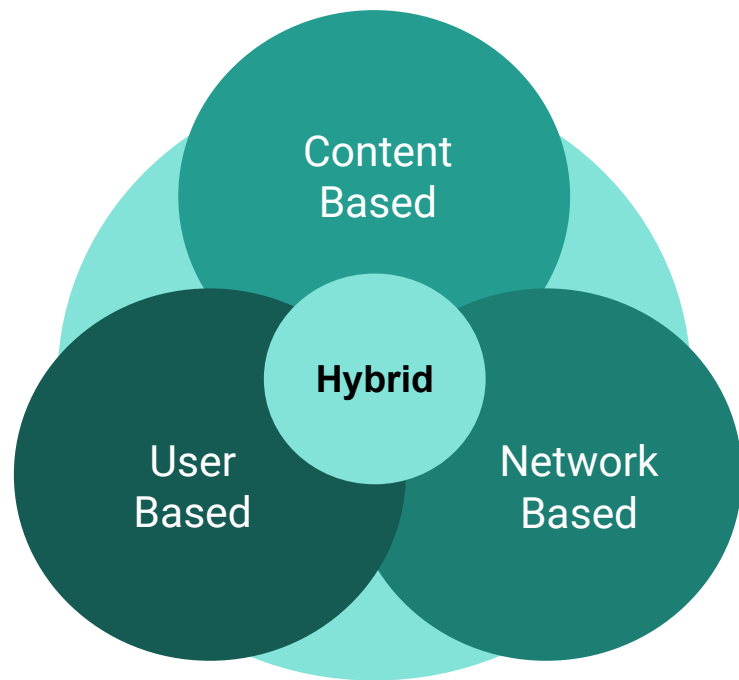
Fake News Detection Methods



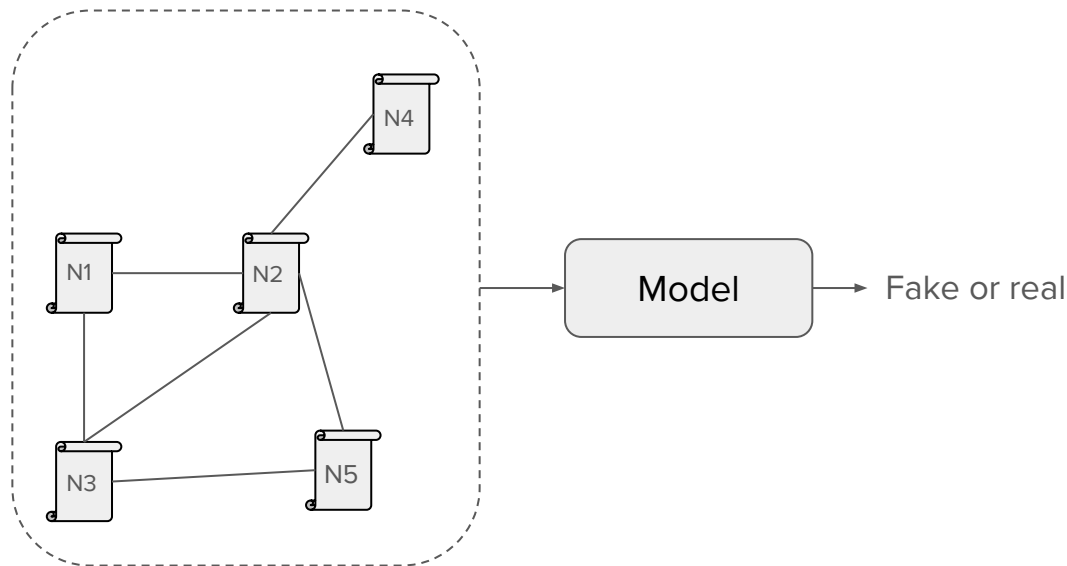
Content Based



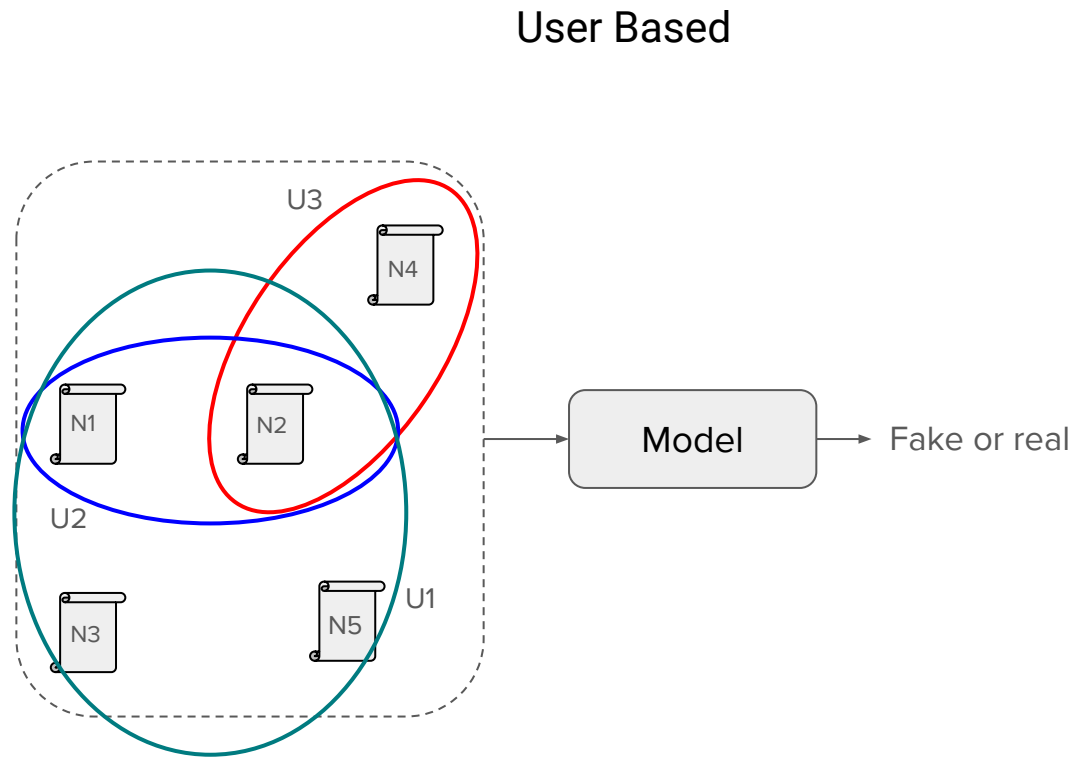
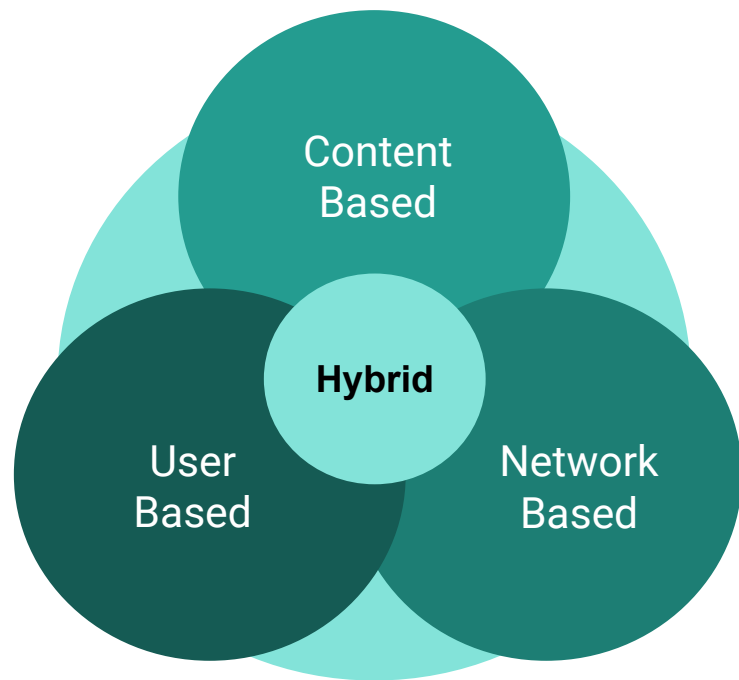
Fake News Detection Methods



Network Based



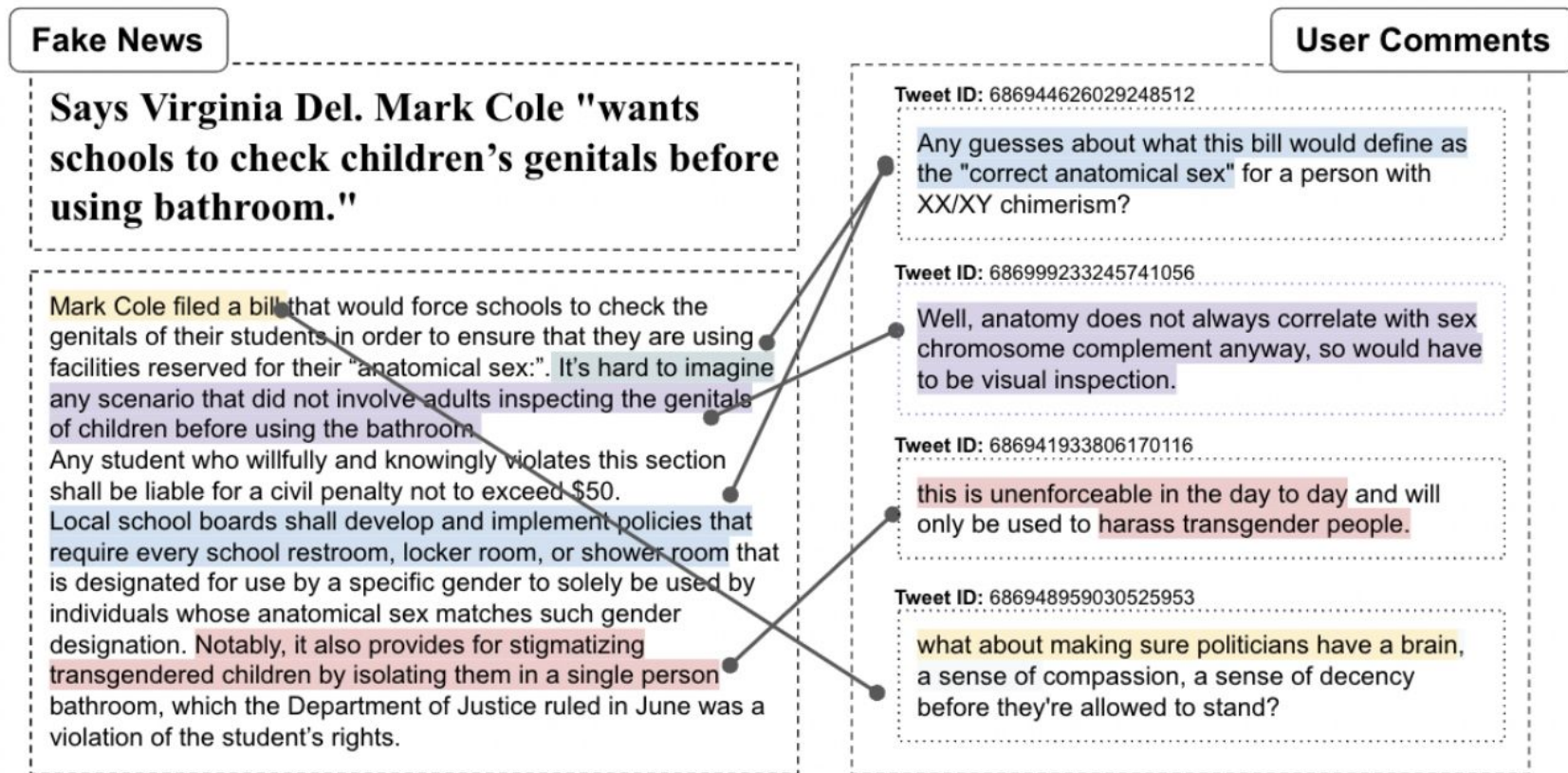
Fake News Detection Methods



Transparency of Fake News Detection Models

- Initially, we utilized some sophisticated ML/DL models for detecting the fake news. (Often got good performance scores as well.)
 - However, predicting only the Fake or Real label may not be always sufficient.
 - In a sense, why should we trust a model prediction, when it's prone to mispredicts the labels?
 - Arguably, even if we get correct predictions, we always look for evidence or justification for something being real or fake.
-
- Fake News Detection with Evidence

Evidence-based Fake-news detection.



Concepts related to Fake News

Concept	Authenticity	Intention
Fake news	Factual	Mislead
Deceptive news	Non-factual	Mislead
Misinformation	Non-factual	Undefined
Claims	Commonly factual	Not bad
Rumors	Undefined	Undefined
Clickbait	Undefined	Mislead

Concepts related to Fake News

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Clickbait	Undefined	Mislead

Claims on Social Media



Manish Tewari
@ManishTewari

CoronaVirus is a bio-weapon that went rouge or was made to go rouge. It is an act of Terror. International investigation conducted either under auspices of ICJ or ICC is necessary to unearth the truth & bring focus back on eradicating Biological Weapons.



@foxnewspolitics
foxnewspolitics

BREAKING NEWS: President @BarackObama assassinated, 2 gunshot wounds have proved too much. It's a sad 4th for #america. #obamadead RIP

8 hours ago via web



Favorite



Retweet



Classic RT



Reply

What is a “claim”?

“State or assert that something is the case, typically without providing evidence or proof.”

- Oxford Dictionary [\[33\]](#)

“A claim is an assertion that deserves our attention”

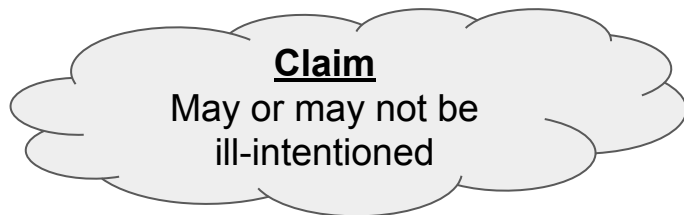
-Toulmin (2003) [\[34\]](#)

“A claim is a disputed statement that we try to support with reasons.”

-Govier (2013) [\[35\]](#)

Fake news and Claims

- Fake news has a strong relationship with claim
 - A news with claim is a worthy candidate for fake news.



Text	Claim	Fake
Alcohol cures corona.	Yes	Yes
Wearing mask can prevent corona.	Yes	No
Lord, please protect my family & the Philippines from the coronavirus.	No	No
If this corona scare doesn't end soon imma have to intervene	No	No

Motivation

Men burnt alive by mob after false WhatsApp messages accuse them of child kidnap



US deaths due to poisoning rise after Trump endorses bleach, disinfectant as COVID-19 treatment



Tweets

stated on January 28, 2020 in a tweet:

Says drinking a bleach solution will prevent you from getting the coronavirus.

Why is detecting claims “so” difficult?



Relatively unexplored
domain

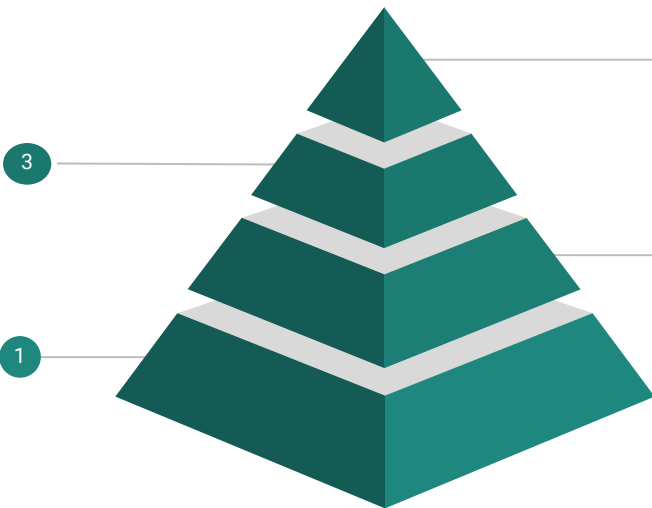
Absence of
unstructured datasets

Lack of sufficient
amount of data

Highly difficult to
differentiate from fake
news and opinions.

Claim Check-worthiness: If the claim is worthy enough to get verified?

Claim Detection: If a post has a claim or not. [Gupta et al., EACL-2021, Sundriyal et al., CIKM - 2021]



Claim Verification: Find evidence for claim and decides its veracity.

Claim Span Identification: What are the claims in the post? [Sundriyal et al., EMNLP-2023]

Problem Statement: Claim Detection



Tweets

RT @PirateAtLaw: No no no. Corona beer is the cure not the disease. Please try this cure. I drink Corona beer a lot!
<https://t.co/fnba2fr2m2>

Claim

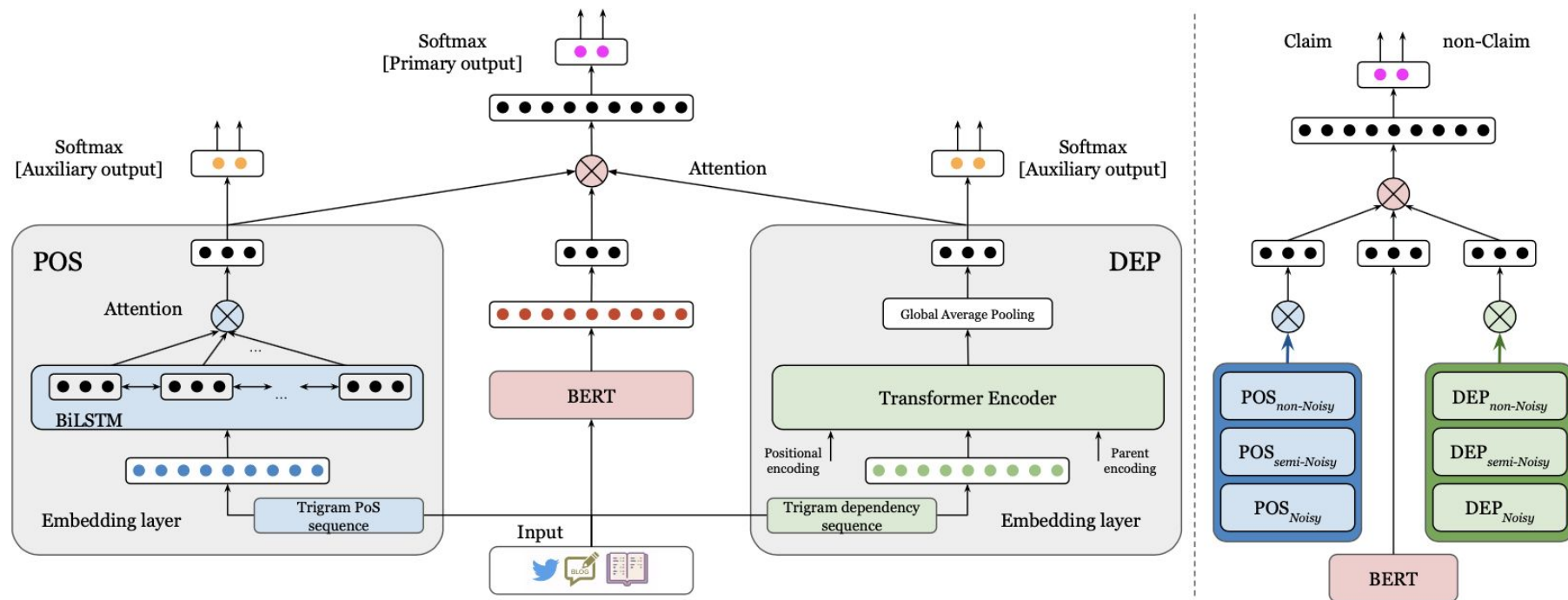


Tweets

Lord, please protect my family & the Philippines from the coronavirus.

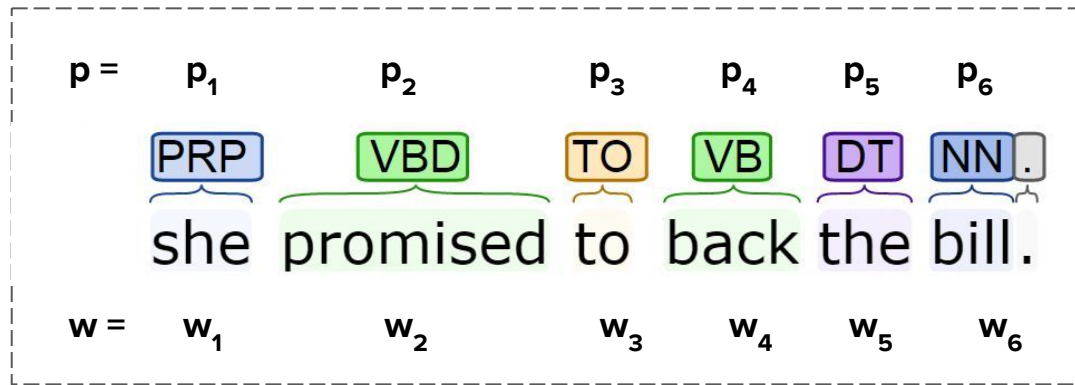
Non-claim

Linguistic Encapsulation and Semantic Amalgamation



1. Part of Speech (POS) Encoding

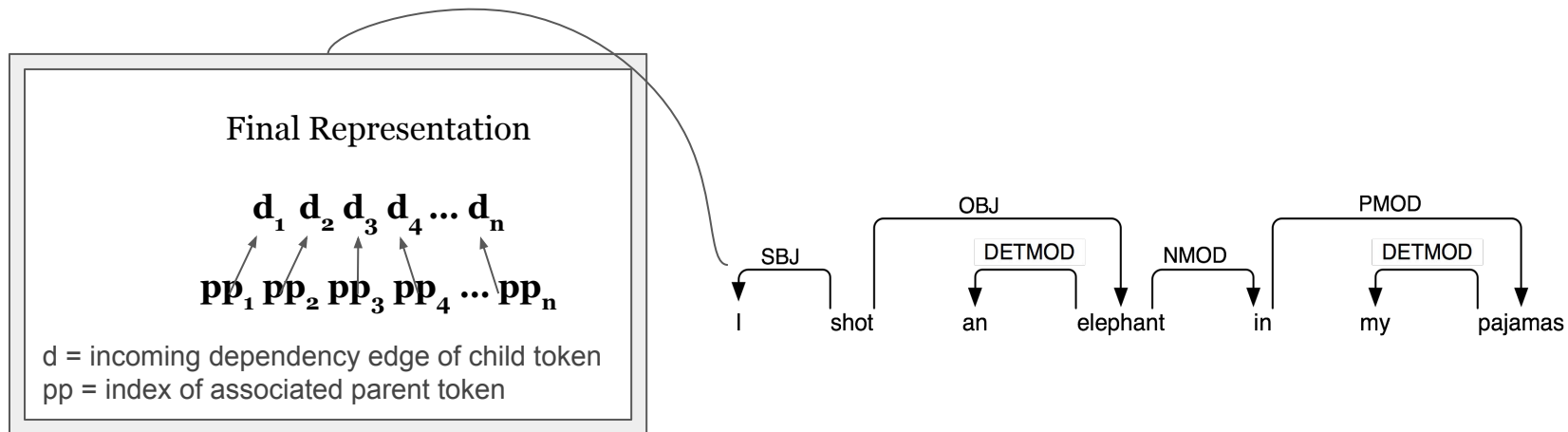
- Text sequence:
 $w_1 w_2 w_3 w_4 \dots w_n$
- Text to POS sequence:
 $p_1 p_2 p_3 p_4 \dots p_n$
- Extract sequence tri-grams:
 $(p_1 p_2 p_3) (p_2 p_3 p_4) \dots (p_{n-1} p_{n-2} p_n)$
- Tri-gram as an individual token:
 $p_i p_{i+1} p_{i+2} = t_i$



2. Dependency (DEP) Tree Encoding

Dep-tag sequence: $\mathbf{d}_1 \mathbf{d}_2 \mathbf{d}_3 \mathbf{d}_4 \dots \mathbf{d}_n$

Dep-tag and Parent pos sequence: $\{(\mathbf{d}_1, \mathbf{pp}_1), \dots (\mathbf{d}_n, \mathbf{pp}_n)\}$

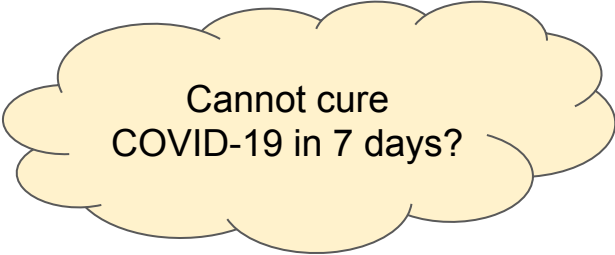


Evaluation

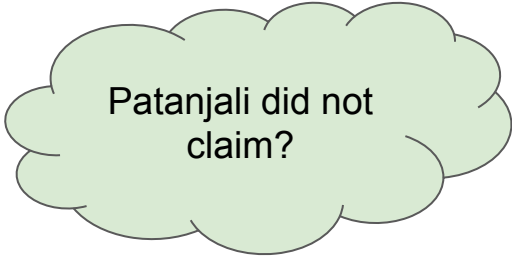
INDUSTRY

Ramdev's Patanjali Ayurved claims it can
"cure COVID-19 in 7 days"

What is the claim?



Cannot cure
COVID-19 in 7 days?



Patanjali did not
claim?

Problem Statement: Claim Span Identification



Tweets

RT @PirateAtLaw: No no no. *Corona beer is the cure not the disease.* Please try this cure. I drink Corona beer a lot!
<https://t.co/fnba2fr2m2>



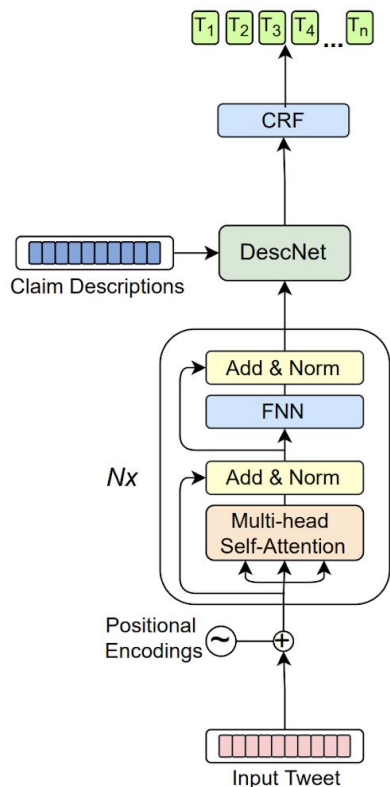
" Corona beer is the cure not the disease. "

BIO encoding for Span

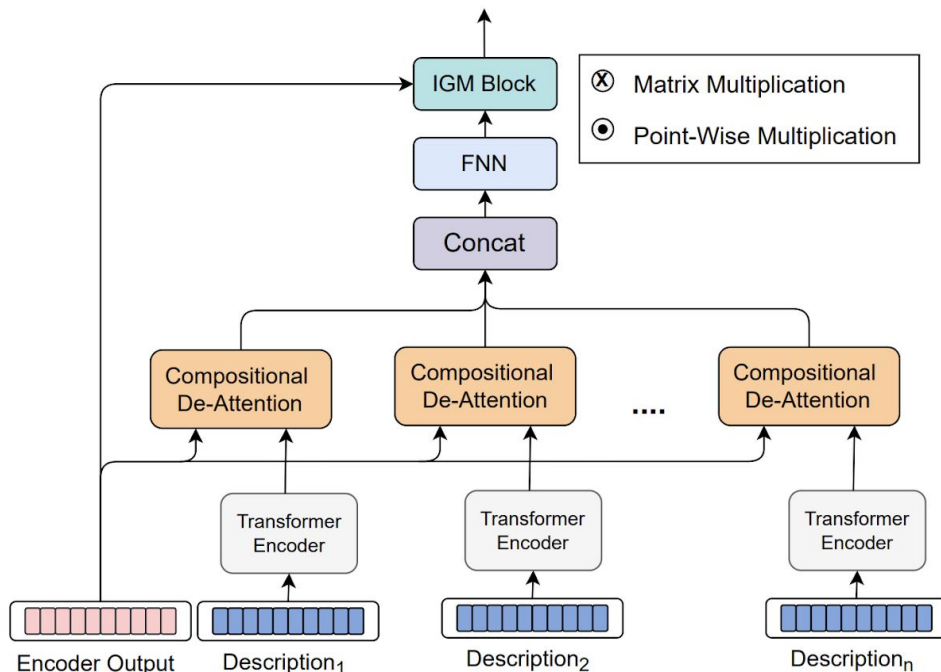
Tweet	Span	BIO Encoding
Alcohol cures corona .	[0,2]	B I I O
Lord , please protect my family & the Philippines from the coronavirus .	-	O O O O O O O O O O O O O O O O O
Heyyaa , it's true . Wearing mask can prevent corona .	[3, 7]	O O O O O B I I I I O
If this corona scare doesn't end soon imma have to intervene .	-	O O O O O O O O O O O O O O O O

- Other Encoding schemes
 - IO (I-inside and O-outside)
 - BIOE (B-beginning, I-inside, E-end, and O-outside)
 - BIESO (B-beginning, I-inside, E-end, S-single word entity and O-outside)

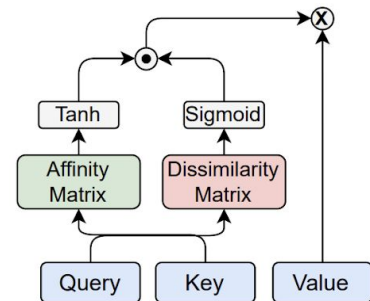
Description **A**ware RoBERTa (DABERTa)



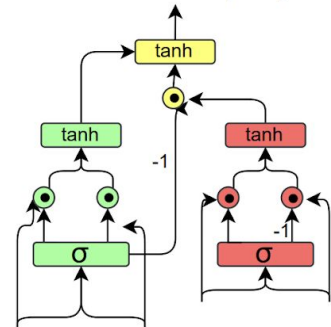
Description Infuser Network (DescNet)



Compositional De-Attention (CoDA)



Interactive Gating Mechanism (IGM)



Claim Descriptions

- Coming from annotation guidelines (manually created document by the domain expert).
 - Definitions of what can be claim and what can't be claim.

Claim Description	Example
Texts in the tweet mentioning statistics, dates or numbers	<i>Another case for more testing for #coronavirus! Blood tests show 14% of people are now immune to covid-19 in one town in Germany.</i>
Texts in the tweet that negate a possibly false claim.	<i>No! #Bleach won't cure #COVID19. Disinfectants can't kill the #coronavirus in your body. In fact, they will hurt you. If you or someone you know has been exposed to bleach, call Poison Control for help (1-800-222-1***).</i>
Texts in the tweet containing opinions that have societal implications.	<i>@username @username I think it's a bio weapon made by China so I'm not surprised it has a lot of carriers.</i>

Evaluation Metrics

- Let's consider the below example,
 - Gold Standard: Anthony Edward Stark
 - Predicted Answer: Edward Stark....
- Evaluation:
 - Free Match: Match the labels (BIO)
 - TP: 2, FP: 0, FN: 1 $\Rightarrow$ Precision: 1, Recall: 0.66
 - Entity Matching
 - Exact Match (Strict): Match the token completely
 - EM = 0
 - Partial Match (Relax): Check overlap with certain threshold.
 - Threshold (60%): PM: 1
 - Threshold (75%): PM: 0

Humans-in-the-loop Analysis

	Model	Tweet
1	Gold	Gold Truly sobering analysis: US more vulnerable than many countries to #coronavirus owing to combination of high numbers of uninsured, many w/o paid sick leave, and a leadership that has downplayed the challenge while not preparing the country for it.
	RoBERTa	Gold Truly sobering analysis: US more vulnerable than many countries to #coronavirus owing to combination of <i>high numbers of uninsured</i> , many w/o <i>paid sick leave</i> , and a leadership that has downplayed the challenge while not preparing the country for it.
	DABERTa	Gold Truly sobering analysis: US more vulnerable than many countries to #coronavirus owing to combination of high numbers of uninsured, many w/o paid sick leave, and a leadership that has downplayed the challenge while not preparing the country for it.
2	Gold	Whether made on purpose or not #coronavirus was used by the #CCP as a bio weapon , not only to kill people but to encourage racism among their citizens against foreigners. Especially black people, CCP is kicking out black people from hotels even if they dont have covid.
	RoBERTa	Whether made on purpose or not #coronavirus was used by the #CCP as a bio weapon , <i>not only to kill people but to encourage racism among their citizens against foreigners</i> . Especially black people, CCP is kicking out black people from hotels even if they dont have covid.
	DABERTa	Whether made on purpose or not #coronavirus was used by the #CCP as a bio weapon , <i>not only to kill people but to encourage racism among their citizens against foreigners</i> . Especially black people, CCP is kicking out black people from hotels even if they dont have covid.
3	Gold	RT @HealtheNews: Can honey, ginger, garlic or turmeric or any other home remedies cure #Covid19? No, here's why.
	RoBERTa	RT @HealtheNews: Can <i>honey, ginger, garlic or turmeric or any other home remedies cure #Covid19?</i> No, here's why.
	DABERTa	RT @HealtheNews: Can <i>honey, ginger, garlic or turmeric or any other home remedies cure #Covid19?</i> No, here's why.

Error analysis of the outputs. Bold text (green) highlights the correct claim span whereas text in italics (red) represents the mistakes committed by our model, DABERTa, and vanilla RoBERTa as baseline.

Problem Statement: Claim Verification



Tweets

It's much more than a coincidence that China has a bioweapons lab with sloppy protocols in Wuhan. Seems like a very bad place to have a bioweapons lab. The whole "this came from snakes" Chinese party line makes me think the virus was manufactured



False claim

But why? Can you justify?

Evidence for false claim from some trustable source:

As the Covid-19 pandemic continues its destructive course, two theories are being widely aired...The lab is one of 20 such facilities under the Chinese Academy of Sciences, but is the only one dealing with virology. Fully compliant with ISO standards, the Wuhan facility interacts regularly with a host of outside experts. Like other labs, its aim is to protect populations against new viruses..



Tweets

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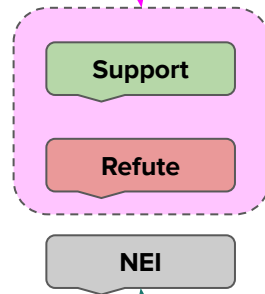
Wikipedia/Internet



Relevant Documents



Evidence



Justified by the evidence!!

Support

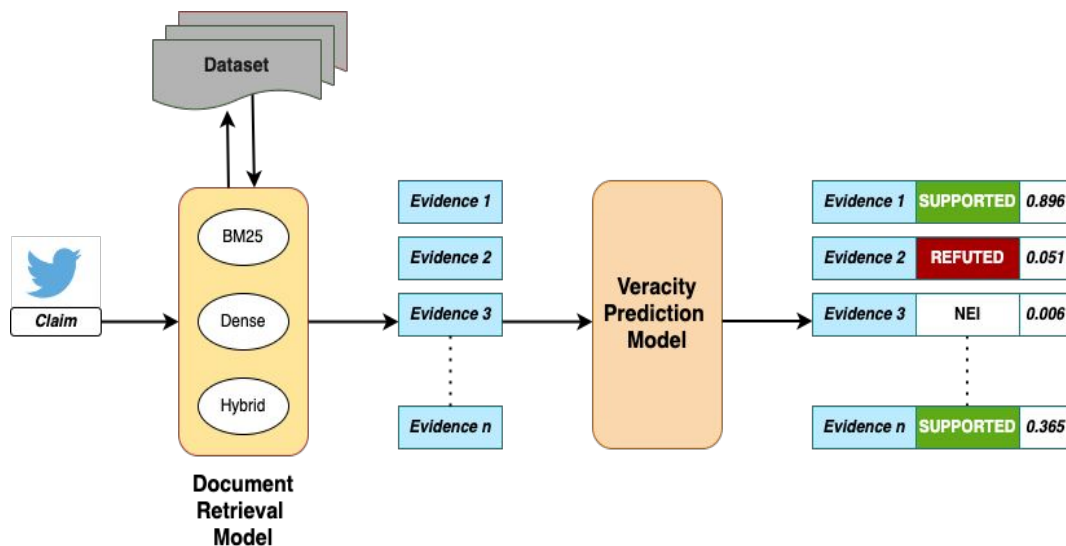
Refute

NEI

Veracity

*Could not find an evidence; hence,
Not-Enough-Information (NEI)*

Document Retrieval and Claim Verification to Mitigate COVID-19 Misinformation

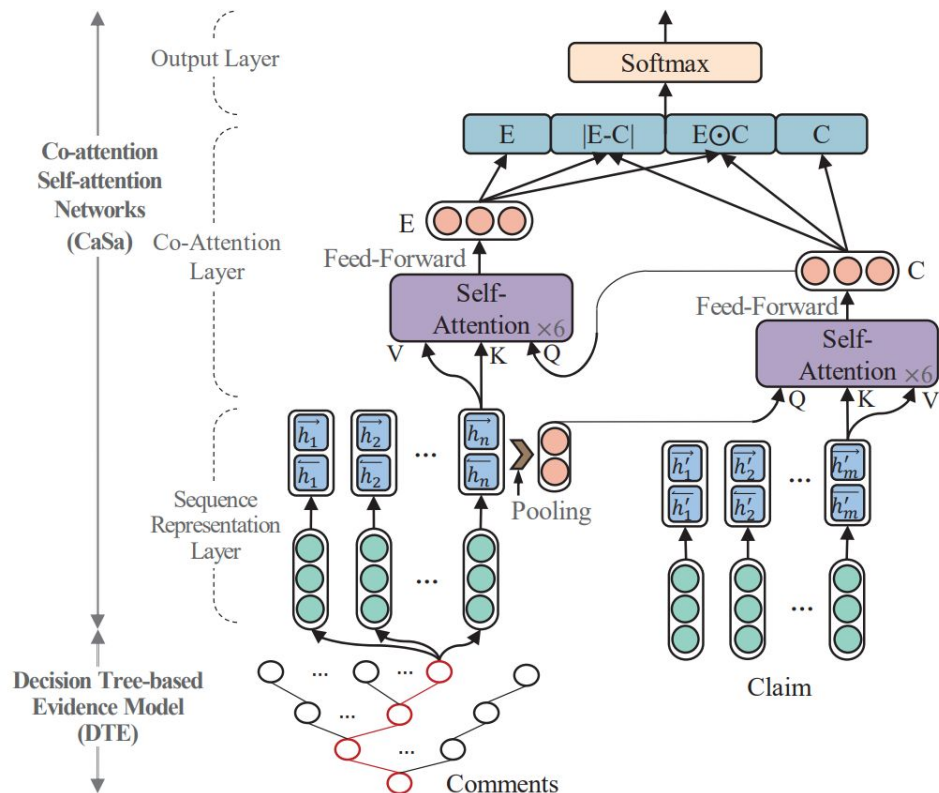


Document retrieval module: Uses one of the given datasets to retrieve top-k relevant documents for the corresponding input claim

Veracity prediction module: Seeks to establish the retrieved documents' credibility against the input claim.

DTCA: Decision Tree-based Co-Attention Networks for Explainable Claim Verification (Wu et. al. 2020)

- **Decision Tree-based Evidence model (DTE)** to select comments with high credibility as evidence in a transparent and interpretable way.
- **Co-attention Self-attention networks (CaSa)** to make the selected evidence interact with claims.



CheckThat! Lab Shared Task

